Kunj Rathod

Applied AI Engineer — Agentic Platforms, RAG/Data Systems, Developer Tooling & On-Device AI
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Education

University of Utah — B.S. Computer Science, GPA 3.7/4.0, Dean's List Aug 2023 – Dec 2026
Coursework: Distributed Systems, Operating Systems, Machine Learning, Computer Vision, NLP, Algorithms, Databases

Experience

Software Engineering Intern — Microsoft Fabric (Azure Data) May 2026 – Aug 2026
Microsoft, Redmond, WA

- Building SKU Change History end-to-end for the Fabric Manage Capacities admin portal: query builder, service layer, REST API, data contracts, and telemetry-backed audit timelines across enterprise tenants.
- Migrating capacity overage workflows into native Fabric platform UX; partnering with UX/backend teams across 3 time zones to turn complex program data into trustworthy, reviewable admin workflows.

Software Development Intern, AI Services (SUDO Program) Jan 2025 – Present
University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant LLM platform for 90+ hospital executives: AWS Bedrock Agents, Knowledge Bases, Guardrails, streaming inference, retrieval-backed context, API caching, and persistent sessions for 1,000+ conversations.
- Reduced latency 40% with p95 time-to-first-token <200ms; owned API design, schema, access-sensitive cloud infrastructure, and 6 production features across React/TypeScript, Flask, Lambda, DynamoDB, S3, and RDS.

AI Engineering Intern Nov 2024 – Apr 2025
CourtEasy.ai / Nugen, Remote

- Scaled hybrid RAG over 10M+ legal documents using dense retrieval, BM25, and reranking; improved recall 28%, reduced hallucinations 35%, and served 5,000+ daily legal-research queries.
- Benchmarked 8 LLM families on accuracy, latency, and cost; built model-routing analysis projected to cut inference spend by $50k+/yr while preserving answer quality.

AI Platforms, Agents & Evaluation

Continuum / FNDR — Local-First Multimodal Agent Memory Engine 2024 – Present
Rust, Tauri 2, React, TypeScript, LanceDB, Apple Vision, OCR, embeddings, MCP

- Built a privacy-first desktop memory layer that captures foreground macOS context on-device into vector + lexical retrieval fields, reconstructs temporal context from screen/audio/web activity, and exposes agent-ready recall through an MCP server.
- Integrated Apple Vision OCR, local Whisper transcription, CLIP-style embeddings, and local/on-device LLM/VLM inference for searchable knowledge over user activity without cloud dependency.

HirePilot — 9-Agent Autonomous Recruiting Pipeline Mar 2026
Podium AI Hackathon Winner; Python/FastAPI, Anthropic, Perplexity, GitHub REST, Twilio, Slack, Google Calendar

- Shipped a real-integration agent workflow in 24 hours: GitHub sourcing, enrichment, Anthropic ranking, live Twilio voice interviews, transcript analysis, Slack approval controls, and calendar scheduling.

MEMROT v2 — Reproducible Evaluation of Long-Context Memory Failure May 2026 – Present
Independent research artifact; Gemma-2-2B, Gemma Scope SAEs, LongMemEval-S, CHPC/SLURM

- Built an ablation-heavy evaluation harness for controlled haystack retrieval: gold-position randomization, in-hook attention reduction, empirical SAE layer scans, mean-ablation/intervention tests, and an 840-pass sweep over 140 questions.
- Separated model behavior from measurement artifacts with explicit limitations around feature selection, attention normalization, provenance, and heuristic grading; prioritized reproducible evidence and feedback loops.

Selected Research

Long-term Active Embodied QA (LA-EQA) Spring 2026
University of Utah; embodied agents, VLM video QA, episodic memory, Qwen3-VL

- Built Video Mind Palace, a sampled-frame VLM baseline replacing Robotic Mind Palace's offline GPT-4o scene-graph pipeline; achieved 31–57% online inference speedup at 13–17% relative accuracy tradeoff.

Technical Skills

Languages: Python, TypeScript, Swift, Rust, C++, Java, SQL **Platforms:** macOS/iOS apps, Tauri, React, FastAPI, Flask, REST APIs, CI/CD
AI Systems: agent orchestration, tool-use/function calling, MCP, RAG/GraphRAG, embeddings, vector search, BM25, reranking, evals/ablations, feedback loops, PyTorch, TensorFlow, VLMs
Cloud/Data: AWS Bedrock, Lambda, API Gateway, S3, DynamoDB, RDS, Azure, Docker, Kubernetes, NebulaGraph, LanceDB, PostgreSQL, Chroma/Pinecone-style vector DBs

Kunj Rathod

AI Systems Engineer — Coding Agents, Agent Harnesses, Evals, Developer Tooling & Reliable LLM Systems
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University of Utah — B.S. Computer Science, GPA 3.7/4.0, Dean's List Aug 2023 – Dec 2026
Coursework: Distributed Systems, Operating Systems, Machine Learning, Computer Vision, NLP, Algorithms, Databases

Experience

Software Engineering Intern — Microsoft Fabric (Azure Data) May 2026 – Aug 2026
Microsoft, Redmond, WA

- Building SKU Change History end-to-end for the Fabric Manage Capacities admin portal: query builder, service layer, REST API, data contracts, and telemetry-backed audit timelines across enterprise tenants.
- Migrating capacity overage workflows into native Fabric platform UX; partnering across UX/backend teams to turn ambiguous product/debugging needs into durable platform behavior and observable admin workflows.

Software Development Intern, AI Services (SUDO Program) Jan 2025 – Present
University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant LLM serving infrastructure for 90+ hospital executives: AWS Bedrock Agents, Knowledge Bases, Guardrails, streaming inference, API caching, and session persistence for 1,000+ conversations.
- Reduced latency 40% with p95 time-to-first-token <200ms; owned API design, schema, observability-minded debugging, and 6 production features across React/TypeScript, Flask, Lambda, DynamoDB, S3, and RDS.

AI Engineering Intern Nov 2024 – Apr 2025
CourtEasy.ai / Nugen, Remote

- Scaled hybrid RAG over 10M+ legal documents using dense retrieval, BM25, and reranking; improved recall 28%, reduced hallucinations 35%, and served 5,000+ daily legal-research queries.
- Benchmarked 8 LLM families on accuracy, latency, and cost; built model-routing analysis projected to cut inference spend by $50k+/yr while preserving answer quality.

Agent Systems, Evals & Research

MEMROT v2 — Mechanistic Evaluation of Long-Context Memory Failure May 2026 – Present
Independent research artifact; Gemma-2-2B, Gemma Scope SAEs, LongMemEval-S, CHPC/SLURM

- Built an ablation-heavy evaluation harness for controlled haystack retrieval: gold-position randomization, in-hook attention reduction, empirical SAE layer scans, mean-ablation/intervention tests, and an 840-pass sweep over 140 exactly-one-gold questions.
- Separated model behavior from measurement artifacts with explicit limitations around feature selection, attention normalization, and heuristic grading; prioritized reproducible negative/causal evidence over inflated claims.

Long-term Active Embodied QA (LA-EQA) Spring 2026
University of Utah; embodied agents, VLM video QA, episodic memory, Qwen3-VL

- Built Video Mind Palace, a sampled-frame VLM baseline replacing Robotic Mind Palace's offline GPT-4o scene-graph pipeline; achieved 31–57% online inference speedup at 13–17% relative accuracy tradeoff.
- Audited failure modes in the 100-question benchmark and proposed SceneSmith + MuJoCo environments for longer, interactive, multi-agent embodied QA evaluation.

Selected Projects

Continuum / FNDR — Local-First Multimodal Agent Memory Engine 2024 – Present
Rust, Tauri 2, React, TypeScript, LanceDB, Apple Vision, OCR, embeddings, MCP

- Built a privacy-first desktop memory layer that captures foreground context on-device into vector + lexical retrieval fields, reconstructs temporal context from screen/audio/web activity, and exposes agent-ready recall through an MCP server.

HirePilot — 9-Agent Autonomous Recruiting Pipeline Mar 2026
Podium AI Hackathon Winner; Python/FastAPI, Anthropic, Perplexity, GitHub REST, Twilio, Slack, Google Calendar

- Shipped a real-integration agent workflow in 24 hours: GitHub sourcing, enrichment, Anthropic ranking, live Twilio voice interviews, transcript analysis, Slack approval controls, and calendar scheduling.

Technical Skills

Languages: Python, Rust, TypeScript, C++, Java, Swift, SQL **Systems:** Docker, Kubernetes, Linux/macOS tooling, REST APIs, CI/CD, SLURM, gRPC
AI Systems: coding agents, agent orchestration, tool-using LLMs, evals/ablations, safe tool workflows, RAG/GraphRAG, vector search, BM25, reranking, MCP, PyTorch, TensorFlow, VLMs
Cloud/Data: AWS Bedrock, Lambda, API Gateway, S3, DynamoDB, RDS, Azure, FastAPI, Flask, NebulaGraph, LanceDB, PostgreSQL

Kunj Rathod

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Education

University of Utah Aug 2023 – Dec 2026
B.S. in Computer Science GPA: 3.7/4.0, Dean's List
Coursework: Distributed Systems, Operating Systems, Machine Learning, Computer Vision, NLP, Algorithms, Databases

Industry Experience

Software Engineering Intern — Microsoft Fabric (Azure Data) May 2026 – Aug 2026
Microsoft, Redmond, WA

- Building the SKU Change History feature end-to-end (query builder, service layer, REST API) for the Fabric Manage Capacities admin portal, integrating internal telemetry pipelines to give capacity admins auditable SKU-change timelines across large enterprise tenants.
- Migrating capacity overage workflows into native Fabric platform UX with action-driven side panes; defining data contracts with UX and backend teams across 3 time zones.

Software Development Intern, AI Services (SUDO Program) Jan 2025 – Present
University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant serving infrastructure for 90+ hospital executives: token-streaming inference at p95 <200ms TTFT and 40% latency reduction via pipeline optimization and API caching.
- Owned API design, schema, and AWS infrastructure (Lambda, API Gateway, RDS, DynamoDB, S3, Bedrock); shipped 6 features across 4 sprints and implemented distributed session persistence for 1,000+ concurrent conversations.

AI Engineering Intern Nov 2024 – Apr 2025
CourtEasy.ai / Nugen (Remote)

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) over 10M+ documents, improving recall 28% and reducing hallucinations 35%; benchmarked 8 model families on F1, latency, and cost and built routing that cut inference spend.

Selected Projects

BioGraphRAG — Distributed Biomedical Knowledge Graph System May 2024 – Jan 2025
Python, FastAPI, NebulaGraph, LlamaIndex, Docker, AWS

- Engineered a distributed GraphRAG system over 1M+ biomedical entities (UniProt, AlphaFold, RXNav) with a 40% accuracy gain over embedding-only baselines; optimized graph traversal 3x via caching and node pruning (p95 <500ms) and built ETL pipelines processing 2M+ entity updates monthly.

Continuum — Local-First Multimodal Memory Engine Jun 2026
Rust, Tauri 2, React, TypeScript, LanceDB, OCR, embeddings, MCP

- Built a privacy-first desktop memory layer capturing foreground context on-device and storing vector + lexical retrieval fields in LanceDB, exposing agent-ready recall through an MCP server.

HirePilot — Autonomous AI Recruiting Pipeline (Podium AI Hackathon Winner) Mar 2026
Python/FastAPI, Anthropic, Perplexity, GitHub REST, Twilio, Slack, Google Calendar

- Shipped a 9-agent recruiting pipeline in 24 hours integrating GitHub sourcing, Perplexity enrichment, live Twilio voice interviews, ranked shortlisting, and manager approval controls.

Research

MEMROT v2 — Mechanistic Evaluation of Long-Context Memory Failure May 2026 – Present
Independent; Gemma-2-2B, Gemma Scope SAEs, LongMemEval-S, CHPC/SLURM

- Built a reproducible mechanistic-evaluation pipeline (gold-position randomization, SAE layer scans, mean-ablation and intervention tests) running an 840-pass sweep over 140 controlled retrieval questions on distributed CHPC/SLURM compute.

Circuit-Level Generalization via Sparse Feature Circuits Jan 2026 – Present
University of Utah; attribution graphs, graph skeletonization

- Open-sourced a graph-skeletonization tool that prunes attribution graphs and groups nodes into supernodes via Mapper embeddings, exporting visualization-compatible artifacts without dropping source metadata.

Languages & Technologies

Languages: Python, C++, Java, Rust, Swift, TypeScript, SQL
Systems & Cloud: AWS (Lambda, API Gateway, S3, DynamoDB, RDS, Bedrock), Azure, Docker, Kubernetes, FastAPI, gRPC, CI/CD, SLURM, distributed systems
Data & AI: RAG/GraphRAG, vector search, BM25, reranking, NebulaGraph, LanceDB, LlamaIndex, PyTorch, TensorFlow, model evaluation

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Software engineer focused on native Apple platform development and privacy-first, on-device intelligence: Swift/SwiftUI apps and Rust systems with Metal-accelerated inference on Apple Silicon.

Education

University of Utah Aug 2023 – Dec 2026
B.S. in Computer Science | GPA: 3.7/4.0, Dean's List *Salt Lake City, UT*
Coursework: Operating Systems, Distributed Systems, Computer Vision, Machine Learning, NLP, Algorithms, Databases.

Apple Platform & On-Device Projects

FNDR – Privacy-First Local AI Assistant for macOS Jan 2024 – Present
Rust, Metal, Apple Vision Framework, ONNX, Llama 3.2, Whisper

- Built a zero-trust, local-only memory assistant for macOS with full data sovereignty: no cloud, no telemetry, all inference and storage on device.
- Optimized on-device LLM (Llama 3.2) and VLM (SmolVLM) inference with Metal-accelerated backends for low-latency RAG on M-series Apple Silicon.
- Architected a real-time screen-extraction pipeline on the Apple Vision Framework for high-speed OCR and CLIP visual embeddings, reconstructing temporal context from screen snapshots.
- Added on-device meeting intelligence with local Whisper transcription and a temporal search engine over vector + lexical fields, modeling semantic links across activity, web, and audio.

Wingman – Multi-Modal AI Personal Assistant (iOS) Apr 2025 – Present
SwiftUI, MVVM, OpenAI API (GPT-4o, Whisper), Firebase

- Built a native iOS assistant in SwiftUI on a clean MVVM architecture, supporting three input modalities in one conversational surface: voice, text chat, and image.
- Integrated GPT-4o for multimodal, context-aware responses and Whisper for voice transcription, grounded by RAG-enhanced memory over prior conversation history.
- Implemented an offline-first design with Firebase sync, real-time streamed message rendering, and persistent conversation history across sessions.

Experience

Software Engineering Intern – Fabric (Azure Data) May 2026 – Aug 2026
Microsoft *Redmond, WA*

- Building the SKU Change History feature end-to-end (query builder, service layer, REST API) for the Fabric Manage Capacities admin portal, with telemetry-backed auditable change timelines across large enterprise tenants.
- Defining data contracts with UX and backend teams across 3 time zones; migrating capacity workflows into native platform UX.

Software Development Intern, AI Services (SUDO Program) Jan 2025 – Present
University of Utah Health *Salt Lake City, UT*

- Architected serving infrastructure for 90+ hospital executives: token-streaming inference at p95 <200ms time-to-first-token and 40% latency reduction via pipeline optimization and caching.
- Owned API design, schema, and cloud infrastructure; shipped 6 features across 4 sprints with distributed session persistence for 1,000+ concurrent conversations.

AI Engineering Intern Nov 2024 – Apr 2025
CourtEasy.ai / Nugen *Remote*

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) over 10M+ documents (+28% recall, –35% hallucinations); benchmarked 8 model families and built routing that cut inference spend.

Research

Mechanistic Interpretability & Embodied Agents 2026 – Present
University of Utah

- MEMROT: reproducible pipeline probing long-context memory failure in Gemma-2-2B via SAE layer scans and mean-ablation interventions on distributed compute.
- Spatial memory for embodied agents: retrieval-augmented spatial memory for long-horizon navigation; evaluated on OS Marathon, BrowseComp, and Mind2Web.

Technical Skills

Apple / Native: Swift, SwiftUI, MVVM, Xcode, Metal, Apple Vision Framework, on-device LLM/VLM inference, Apple Silicon
Languages: Swift, Rust, C++, Python, TypeScript, Java, SQL
Systems & ML: ONNX, PyTorch, RAG/GraphRAG, LanceDB, MCP, FastAPI, Docker, AWS (Bedrock, Lambda), CI/CD

Kunj Rathod

AI Research Engineer — Embodied AI, LLM Systems & Agents

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Education

B.S. Computer Science — GPA 3.7/4.0, Dean's List *Aug 2023 – Dec 2026*
University of Utah Salt Lake City, UT

- Relevant: Machine Learning, Computer Vision, NLP, Distributed Systems, Interpretability of LLMs, Multimodal LLM Agents

Research

Spatial Memory for Embodied & Long-Horizon Web Agents *2026 – Present*
University of Utah

- Designing retrieval-augmented memory architectures that unify spatial, temporal, episodic, and semantic memory for long-horizon robot navigation and manipulation (inspired by ReMEmbR / NaVQA).
- Evaluating web agent benchmarks: OS Marathon, BrowseComp, Mind2Web; studying Web Explorer and Web Sailor V2 for trajectory-based multi-turn interaction and extended agentic reasoning.
- Studying SceneSmith (MIT CSAIL / Toyota Research) for procedural generation of simulation-ready environments from natural language for scalable robot policy evaluation.

Long-term Active Embodied QA (LA-EQA) *Spring 2026*
University of Utah | w/ Nikesh Subedi

- Co-authored *"Exploring Long-term Active Embodied QA in Simulated Indoor Environments."* Proposed **Video Mind Palace (VMP)**: replaced Robotic Mind Palace's offline GPT-4o scene-graph pipeline with direct Qwen3-VL video queries, achieving **31–57% inference speedup** at 13–17% relative accuracy drop on the 100-question LA-EQA benchmark (single A100 GPU, Qwen3-VL 8B/30B/FP8).
- Identified three structural benchmark limitations and proposed future direction using SceneSmith + MuJoCo for interactive, multi-agent, long-horizon embodied QA evaluation.

Mechanistic Interpretability — Circuit Generalization in LLMs *2026 – Present*
University of Utah, advised by Ana Marasovic

- Extracting sparse feature circuits from `gemma-2-2b` (GemmaScope transcoders) to predict generalization from default to counterfactual task distributions; built automated attribution graph pipelines (Pathways Discovery).
- Key negative result: suppressing over-refusal SAE features worsened over-refusal rates — publishable finding with implications for circuit-based behavior steering.

Multi-Agent Materials Discovery — STARS Lab *Aug 2025 – Feb 2026*
University of Utah | NASA, Microsoft, U.S. DoD collaboration Salt Lake City, UT

- Built graph-augmented multi-agent pipeline extracting material properties from 1,000+ papers for automated Ashby plot generation; developed constraint-based design region engine for aerospace materials selection.

Experience

Software Engineer Intern — Azure Data *May 2026 – Present*
Microsoft Redmond, WA

- Building distributed data systems on Azure; full-stack engineering across the Azure Data ecosystem.

AI Services Intern — SUDO Program *Jan 2025 – Present*
University of Utah Health Salt Lake City, UT

- Built HIPAA-compliant LLM platform (AWS Bedrock Agents, Knowledge Bases, Guardrails) for 90+ hospital executives; p95 TTFT <200ms, 40% latency reduction, 1,000+ concurrent sessions via DynamoDB+S3 persistence.
- Shipped 6 full-stack features (React/TypeScript, Flask, Lambda) across 4 sprints; owned API design, schema, and infra.

AI Engineering Intern *Nov 2024 – Apr 2025*
CourtEasy.ai / Nugen Remote

- Hybrid RAG over 10M+ legal documents (dense + BM25 + reranking); +28% retrieval accuracy, -35% hallucination rate, 5,000+ daily queries.
- Benchmarked 8 LLM families on LegalBench, NyayaAnumana, IL-TUR; routing analysis saved $50k+/yr projected inference cost.

AI Research Intern — BioGraphRAG *May 2024 – Aug 2024*
Garje Marathi Global Salt Lake City, UT

- Distributed GraphRAG over 1M+ biomedical entities (NebulaGraph); +40% factual accuracy, 3× traversal speedup to sub-500ms p95; ETL processing 2M+ entity updates/month. Article republished by NebulaGraph.

Selected Projects

FNDR — Local AI Memory Engine (macOS) *2024 – Present*

Rust, Tauri, Metal, ONNX, Llama 3.2, Whisper, Apple Vision, MCP

- Metal-accelerated on-device LLM/VLM inference, Apple Vision OCR, CLIP embeddings, Graphiti-style temporal knowledge graph for multi-hop reasoning, local Whisper transcription, MCP server for agent interop. Zero cloud dependency.

HirePilot — 9-Agent Autonomous Recruiting System [Podium AI Hackathon — 1st Place] *2026*

TypeScript, Anthropic API, Twilio, Google Calendar, Slack

- End-to-end autonomous hiring: GitHub sourcing, live voice interviews (Twilio + transcript analysis), multi-dimensional scoring, Slack approval loop; replaced full human-in-loop recruiting workflow.

Writing & Publications

- **Exploring LA-EQA in Simulated Indoor Environments** — Rathod, Subedi. Class report, CS Advanced AI, U of Utah, Spring 2026.
- **BioGraphRAG** — co-authored; republished by NebulaGraph. Substack, Oct 2024.
- **LLM Families on Legal Benchmarks** — internal analysis, CourtEasy.ai / Nugen, 2025.

Technical Skills

Languages	Python, C++, Java, TypeScript, Rust, SQL
AI / ML	PyTorch, RAG, GraphRAG, Sparse Autoencoders, LangChain, LlamaIndex, Transformers, Multi-Agent Systems, Embodied AI, VLMs, Web Agents
Infra	AWS Bedrock, Lambda, S3, DynamoDB, Docker, Kubernetes, PostgreSQL, NebulaGraph, ChromaDB, Pinecone

Awards

Kahlert Impact Prize ($1,000) — Kahlert School of Computing, U of Utah, Mar 2026 · **Dean's List** 2024–25 · **Minute0 Hackathon Winner** Feb 2026

Kunj Rathod

AI Researcher — Mechanistic Interpretability & LLM Systems

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Education

B.S. Computer Science — GPA 3.7/4.0, Dean's List *Aug 2023 – Dec 2026*
University of Utah Salt Lake City, UT

- Relevant: Machine Learning, NLP, Interpretability of LLMs, Distributed Systems, Multimodal LLM Agents

Research

Mechanistic Interpretability — Circuit-Level Generalization in LLMs *2026 – Present*
University of Utah, advised by Ana Marasovic | w/ Emma Brown, Christina Dong

- Extracted sparse feature circuits from `gemma-2-2b` using GemmaScope cross-layer transcoders (Wu et al. framework) to predict whether circuits learned on default tasks generalize to counterfactual task variants.
- Built automated attribution graph pipelines using Pathways Discovery (PD); synthesized prior work across 15+ interpretability papers on circuit stability, feature reuse, and generalization correlation.
- **Key finding:** suppressing over-refusal SAE features unexpectedly worsened over-refusal rates rather than reducing them — a publishable negative result with direct implications for activation-steering-based safety interventions.
- Prepared NeurIPS-style paper; research poster presented April 2026.

Long-term Active Embodied QA (LA-EQA) *Spring 2026*
University of Utah | w/ Nikesh Subedi

- Co-authored *"Exploring Long-term Active Embodied QA in Simulated Indoor Environments."* Proposed **Video Mind Palace (VMP)**: replaced Robotic Mind Palace's offline GPT-4o scene-graph pipeline with direct Qwen3-VL video queries, achieving **31–57% inference speedup** at 13–17% relative accuracy drop (100-question benchmark, A100 GPU).

Spatial Memory for Embodied & Long-Horizon Web Agents *2026 – Present*
University of Utah

- Designing retrieval-augmented memory architectures unifying spatial, temporal, episodic, and semantic memory; evaluating on OS Marathon, BrowseComp, Mind2Web benchmarks.

Experience

Software Engineer Intern — Azure Data *May 2026 – Present*
Microsoft Redmond, WA

- Building scalable distributed data systems on Azure; full-stack engineering within the Azure Data ecosystem.

AI Services Intern — SUDO Program *Jan 2025 – Present*
University of Utah Health Salt Lake City, UT

- Built HIPAA-compliant LLM platform serving 90+ hospital executives via AWS Bedrock Agents, Knowledge Bases, and Guardrails; achieved p95 TTFT <200ms and 40% latency reduction with DynamoDB+S3 session persistence at 1,000+ concurrent conversations.
- Owned API design, schema, UI (React/TypeScript), and Lambda orchestration across 6 features shipped in 4 sprints.

AI Engineering Intern — Legal AI *Nov 2024 – Apr 2025*
CourtEasy.ai / Nugen Remote

- Benchmarked 8 LLM families (InLegalBERT, InLegalLLaMA, GPT-4o-mini) on LegalBench, NyayaAnumana, IL-TUR, LegalBench-RAG across F1, latency, cost; analysis cut projected inference spend $50k+/yr.
- Built hybrid RAG (dense + BM25 + reranking) over 10M+ legal documents; +28% retrieval accuracy, -35% hallucination rate.

AI Research Intern — BioGraphRAG *May 2024 – Aug 2024*
Garje Marathi Global Salt Lake City, UT

- GraphRAG over 1M+ biomedical entities (UniProt, AlphaFold, RXNav) on NebulaGraph; +40% factual accuracy, 3× graph traversal speedup to sub-500ms p95. Article republished by NebulaGraph.

Selected Projects

FNDR — Privacy-First Local AI Memory (macOS) *2024 – Present*
Rust, Tauri, Metal, ONNX, Llama 3.2, Whisper, Apple Vision, MCP

- Zero-trust local RAG assistant: Metal-accelerated on-device LLM/VLM inference, Apple Vision OCR, CLIP embeddings, Graphiti-style temporal knowledge graph, local Whisper transcription, MCP server for agent interop.

HirePilot — 9-Agent Autonomous Recruiting System [Podium AI Hackathon — 1st Place] *2026*
TypeScript, Anthropic API, Twilio, Google Calendar, Slack

- End-to-end autonomous hiring pipeline: GitHub sourcing, live voice interviews via Twilio + transcript analysis, multi-dimensional scoring, manager approval loop via Slack.

Writing & Publications

- **Exploring LA-EQA in Simulated Indoor Environments** — Rathod, Subedi. Class report, CS Advanced AI, U of Utah, Spring 2026.
- **BioGraphRAG** — co-authored technical article; republished by NebulaGraph. Substack, Oct 2024.
- **LLM Families on Legal Benchmarks** — internal comparative analysis, CourtEasy.ai / Nugen, 2025.

Technical Skills

Languages	Python, C++, Java, TypeScript, Rust, SQL
AI / ML	PyTorch, Sparse Autoencoders, Attribution Graphs, RAG, GraphRAG, LangChain, LlamaIndex, Transformers, Multi-Agent Systems, Embodied AI, VLMs
Infra	AWS Bedrock, Lambda, S3, DynamoDB, Docker, PostgreSQL, NebulaGraph, ChromaDB, Pinecone

Awards

Kahlert Impact Prize ($1,000) — Kahlert School of Computing, U of Utah, Mar 2026 · **Dean's List** 2024–25 · **Minute0 Hackathon Winner** Feb 2026

Kunj Rathod

AI Engineer — Embodied AI, Robotics Systems & Low-Latency Inference

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Education

B.S. Computer Science — GPA 3.7/4.0, Dean's List — *Aug 2023 – Dec 2026*
University of Utah — Salt Lake City, UT

- Relevant: Machine Learning, Computer Vision, NLP, Distributed Systems, Algorithms, Multimodal LLM Agents

Research

Spatial Memory for Embodied & Long-Horizon Web Agents — *2026 – Present*
University of Utah

- Designing retrieval-augmented memory architectures (spatial, temporal, episodic, semantic) for long-horizon robot navigation and manipulation; inspired by ReMEmbR / NaVQA paradigm.
- Studying SceneSmith (MIT CSAIL / Toyota Research) for procedural generation of physics-ready simulation environments, enabling scalable robotic policy evaluation without manual scene authoring.

Long-term Active Embodied QA (LA-EQA) — *Spring 2026*
University of Utah | w/ Nikesh Subedi

- Co-authored *"Exploring Long-term Active Embodied QA in Simulated Indoor Environments."* Proposed **Video Mind Palace (VMP)**: dropped Robotic Mind Palace's expensive scene-graph pipeline in favor of direct VLM video queries, achieving **31–57% inference speedup** with only 13–17% relative accuracy cost on the full 100-question benchmark (A100, Qwen3-VL 8B/30B/FP8).
- Proposed benchmark evolution using SceneSmith + MuJoCo for interactive multi-agent, long-horizon embodied QA including object manipulation and intent attribution tasks.

Multi-Agent Materials Discovery — STARS Lab — *Aug 2025 – Feb 2026*
University of Utah | NASA, Microsoft, U.S. DoD — Salt Lake City, UT

- Graph-augmented multi-agent pipeline ingesting 1,000+ aerospace materials papers; constraint-based design region engine (temperature, creep, pressure) for automated Ashby plot generation and materials selection for rocket engines and hypersonics.

Experience

Software Engineer Intern — Azure Data — *May 2026 – Present*
Microsoft — Redmond, WA

- Distributed data systems engineering on Azure; full-stack development within the Azure Data platform.

AI Services Intern — SUDO Program — *Jan 2025 – Present*
University of Utah Health — Salt Lake City, UT

- **Inference pipeline:** p95 TTFT <200ms on AWS Bedrock with 40% latency reduction via pipeline optimization, API caching, DynamoDB+S3 session persistence for 1,000+ concurrent LLM conversations.
- Built event-driven Lambda orchestration (React/TypeScript frontend, Flask middleware) for a HIPAA-compliant platform serving 90+ hospital executives; shipped 6 features in 4 sprints.

AI Engineering Intern — *Nov 2024 – Apr 2025*
CourtEasy.ai / Nugen — Remote

- Production hybrid RAG over 10M+ legal documents at 5,000+ daily queries; +28% retrieval accuracy, -35% hallucination rate.
- Benchmarked 8 LLM families (F1, latency, cost) across 4 legal benchmarks; routing decisions saved $50k+/yr projected inference spend.

AI Research Intern — BioGraphRAG — *May 2024 – Aug 2024*
Garje Marathi Global — Salt Lake City, UT

- Distributed GraphRAG over 1M+ biomedical entities on NebulaGraph; 3× graph traversal speedup to sub-500ms p95 via caching and high-degree node pruning; 2M+ entity ETL updates/month.

Selected Projects

FNDR — Local AI Memory Engine (macOS) — *2024 – Present*
Rust, Tauri, Metal/ONNX, Llama 3.2, Whisper, Apple Vision, MCP

- **Edge AI:** Metal-accelerated on-device LLM (Llama 3.2) and VLM (SmolVLM) inference on Apple Silicon; real-time Apple Vision Framework OCR pipeline with CLIP visual embeddings for temporal scene reconstruction.
- Graphiti-style temporal knowledge graph for multi-hop reasoning across screen, audio, and web activity; local Whisper transcription; MCP server for agent interop. Zero cloud, zero telemetry.

FlowVia — V2X Urban Mobility Optimization System — *Apr 2024*

Python, TensorFlow, LSTM, V2X (DSRC/C-V2X), OBD-II

- Vehicle-to-Everything traffic platform: real-time speed recommendations via live SPaT data, LSTM traffic flow prediction (continuously retrained), AES-256 edge-encrypted V2V/V2I/V2N communication.
- Full system: in-vehicle OBD-II unit, mobile interface, cloud ML backend, roadside V2X unit — ultra-low-latency, edge-first architecture.

HirePilot — 9-Agent Recruiting System [Podium AI Hackathon — 1st Place] *2026*
TypeScript, Anthropic API, Twilio, Google Calendar, Slack

- Autonomous multi-agent pipeline: GitHub sourcing, live voice AI interviews (Twilio + transcript analysis), multi-dimensional scoring, automated calendar + Slack approval workflows.

Writing & Publications

- **Exploring LA-EQA in Simulated Indoor Environments** — Rathod, Subedi. Class report, CS Advanced AI, U of Utah, Spring 2026.
- **BioGraphRAG** — co-authored; republished by NebulaGraph. Substack, Oct 2024.
- **FlowVia: Technical Deep Dive** — Substack, Apr 2024.

Technical Skills

Languages	Python, C++, Rust, Java, TypeScript, SQL
AI / ML	PyTorch, TensorFlow, ONNX, RAG, GraphRAG, LangChain, LlamaIndex, Transformers, Multi-Agent Systems, Embodied AI, VLMs, RL (DQN, PPO)
Systems	Metal (Apple Silicon), Edge Inference, V2X Protocols (DSRC/C-V2X), AWS Bedrock, Lambda, DynamoDB, Docker, Kubernetes, NebulaGraph

Awards

Kahlert Impact Prize ($1,000) — Kahlert School of Computing, U of Utah, Mar 2026 · **Dean's List** 2024–25 · **Minute0 Hackathon Winner** Feb 2026

Kunj Rathod

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Education

University of Utah Aug 2023 – Dec 2026
B.S. in Computer Science GPA: 3.7/4.0, Dean's List
Coursework: Distributed Systems, Operating Systems, Machine Learning, Computer Vision, NLP, Algorithms, Databases

Research

MEMROT v2 — Mechanistic Anatomy of Long-Context Memory Failure May 2026 – Present
Independent research artifact; Gemma-2-2B, Gemma Scope SAEs, LongMemEval-S, CHPC/SLURM

- Built a reproducible mechanistic-evaluation pipeline for controlled haystack retrieval at fill levels $k \in \{0, 1, 2, 4, 8, 14\}$: gold-position randomization, in-hook attention reduction, empirical SAE layer scans, mean-ablation/intervention tests, and an 840-pass sweep over 140 exactly-one-gold questions.
- Framed the paper as a single-model case study with explicit limitations around feature selection, attention normalization, and heuristic grading, prioritizing honest failure analysis over inflated claims.

Disentangling Correct Refusal from Over-Refusal via Sparse Feature Interventions Mar 2026 – May 2026
Qwen2.5-0.5B-Instruct; sparse autoencoders; refusal-safety evaluation

- Built a 400-example unsafe/benign-overlap benchmark and trained sparse autoencoders on residual activations to rank refusal-feature families; causal steering of one family improved held-out unsafe refusal from 0.80 to 0.85, while suppressing top over-refusal features did not reliably improve benign compliance — a clean negative result.

Circuit-Level Generalization in LLMs via Sparse Feature Circuits Jan 2026 – Present
University of Utah; attribution graphs, sparse feature circuits, graph skeletonization

- Building automated pipelines to extract sparse feature circuits and predict transfer to non-default tasks via attribution graphs and Pathways Discovery; open-sourced a skeletonization tool that prunes graphs into supernodes through Mapper embeddings while preserving source metadata.

Long-term Active Embodied Question Answering in Simulated Indoor Environments Spring 2026
University of Utah; embodied AI, VLM video QA, episodic memory, SceneSmith, MuJoCo

- Built Video Mind Palace, a sampled-frame VLM baseline for LA-EQA that replaces scene-graph memory; cut online inference time by 31–57% versus Robotic Mind Palace with a 13–17% relative accuracy tradeoff, then proposed SceneSmith + MuJoCo environments for longer interactive embodied QA.

Industry Experience

Software Engineering Intern — Microsoft Fabric (Azure Data) May 2026 – Aug 2026
Microsoft, Redmond, WA

- Building SKU Change History end-to-end (query builder, service layer, REST API) for the Fabric Manage Capacities admin portal with telemetry-backed audit timelines; using Microsoft's internal AI stack to accelerate debugging and implementation planning.

Software Development Intern, AI Services (SUDO Program) Jan 2025 – Present
University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant LLM serving infrastructure for 90+ hospital executives: token-streaming inference at p95 <200ms TTFT and 40% latency reduction; owned AWS Bedrock Agents, Knowledge Bases, and Guardrails plus session persistence for 1,000+ concurrent conversations.

AI Engineering Intern Nov 2024 – Apr 2025
CourtEasy.ai / Nugen (Remote)

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) over 10M+ documents (+28% recall, -35% hallucinations); benchmarked 8 LLM families on F1, latency, and cost and built model routing that cut inference spend.

Selected Projects

Continuum — Local-First Multimodal Memory Engine Jun 2026
Rust, Tauri 2, React, TypeScript, LanceDB, OCR, embeddings, MCP

- Built a privacy-first desktop memory layer capturing foreground context on-device into vector + lexical retrieval fields in LanceDB, exposing agent-ready recall through an MCP server.

HirePilot — Autonomous AI Recruiting Pipeline (Podium AI Hackathon Winner) Mar 2026
Python/FastAPI, Anthropic, Perplexity, GitHub REST, Twilio, Slack, Google Calendar

- Shipped a real-integration 9-agent recruiting loop in 24 hours: GitHub sourcing, Perplexity enrichment, Anthropic ranking and outreach, Twilio live voice interviews, and manager approval controls.

Languages & Technologies

AI/ML: PyTorch, TensorFlow, JAX, vision-language models, video QA, embodied AI, agent evaluation, RAG/GraphRAG, LangChain, LlamaIndex
Systems & Cloud: AWS (Lambda, API Gateway, S3, DynamoDB, RDS, Bedrock), Azure, Docker, Kubernetes, FastAPI, gRPC, CI/CD, SLURM, distributed systems
Languages: Python, C++, Java, Rust, Swift, TypeScript, SQL

Kunj Rathod

Machine Learning Engineer – Production ML, Retrieval Systems, Model Evaluation

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Education

University of Utah Aug 2023 – Dec 2026
B.S. in Computer Science GPA: 3.7/4.0, Dean's List
Coursework: Machine Learning, Distributed Systems, Operating Systems, Computer Vision, NLP, Algorithms, Databases

Industry Experience

Software Engineering Intern – Microsoft Fabric (Azure Data) May 2026 – Aug 2026
Microsoft, Redmond, WA

- Building the SKU Change History feature end-to-end (query builder, service layer, REST API) for the Fabric Manage Capacities admin portal, integrating internal telemetry pipelines to give capacity admins auditable SKU-change timelines across large enterprise tenants.
- Migrating capacity overage workflows into native Fabric platform UX with action-driven side panes; defining data contracts with UX and backend teams across 3 time zones.

Software Development Intern, AI Services (SUDO Program) Jan 2025 – Present
University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant LLM serving infrastructure for 90+ hospital executives: token-streaming inference at p95 <200ms TTFT and 40% latency reduction via pipeline optimization, API caching, and AWS Bedrock-backed model orchestration.
- Owned production APIs, schemas, dashboards-adjacent usage telemetry, and AWS infrastructure (Lambda, API Gateway, RDS, DynamoDB, S3); shipped 6 features across 4 sprints and implemented distributed session persistence for 1,000+ concurrent conversations.

AI Engineering Intern Nov 2024 – Apr 2025
CourtEasy.ai / Nugen (Remote)

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) over 10M+ documents, improving recall 28% and reducing hallucinations 35%; evaluated model quality with F1, latency, cost, and hallucination-rate metrics.
- Benchmarked 8 model families across LegalBench, NyayaAnumana, IL-TUR, and LegalBench-RAG; built model-routing analysis that reduced projected inference spend by $50k+/yr while preserving answer quality.

Selected ML Systems & Research

BioGraphRAG – Distributed Biomedical Knowledge Graph System May 2024 – Jan 2025
Python, FastAPI, NebulaGraph, LlamaIndex, Docker, AWS

- Engineered a distributed GraphRAG system over 1M+ biomedical entities (UniProt, AlphaFold, RXNav) with a 40% accuracy gain over embedding-only baselines; optimized graph traversal 3x via caching and node pruning (p95 <500ms) and built ETL pipelines processing 2M+ entity updates monthly.

MEMROT v2 – Mechanistic Evaluation of Long-Context Memory Failure May 2026 – Present
Independent; PyTorch, Gemma-2-2B, Gemma Scope SAEs, LongMemEval-S, CHPC/SLURM

- Built a reproducible model-evaluation pipeline with gold-position randomization, counterfactual retrieval variants, SAE layer scans, mean-ablation, and intervention tests; ran an 840-pass sweep over 140 controlled retrieval questions on distributed CHPC/SLURM compute.

Circuit-Level Generalization via Sparse Feature Circuits Jan 2026 – Present
University of Utah; PyTorch, attribution graphs, graph skeletonization

- Studying whether sparse feature circuits learned on default tasks generalize to counterfactual task variants; open-sourced a graph-skeletonization tool that prunes attribution graphs and exports visualization-compatible artifacts without dropping source metadata.

Continuum – Local-First Multimodal Memory Engine Jun 2026
Rust, Tauri 2, React, TypeScript, LanceDB, OCR, embeddings, MCP

- Built a privacy-first desktop memory layer capturing foreground context on-device and storing vector + lexical retrieval fields in LanceDB, exposing agent-ready recall through an MCP server.

Languages & Technologies

Languages: Python, Java, C++, Rust, TypeScript, SQL
ML: PyTorch, TensorFlow, transformers, model evaluation, offline metrics, retrieval/ranking, RAG/GraphRAG, vector search, BM25, reranking, embeddings, sparse autoencoders
Production & Data: AWS, Azure, Docker, Kubernetes, FastAPI, REST APIs, distributed systems, telemetry pipelines, SLURM, NebulaGraph, LanceDB, PostgreSQL